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# St. Joseph's College of Arts & Science (Autonomous)
## St. Joseph's College Road, Cuddalore – 607001
## M.Sc Degree Examination - APRIL 2021

**PCH1015T – REACTION KINETICS, ELECTRODE KINETICS AND PHOTO CHEMISTRY**

**Time : 3 hrs** **Max Marks :75**

**SECTION – A (15X1=15)**

Answer **ALL** Questions

**I. Choose the correct answer**

1. According to Collision theory, particles must
   a. Collide every where b. Collide with correct orientation
   c. Must be kept under immense pressures d. Shall not be below their melting points
2. If temperature of a reaction is increased, average kinetic energy will be
   a. Increased b. Decreased c. Constant d. Equal
3. Which of the following substances act as catalysis in the body?
   a. Carbohydrates b. Lipids c. Nucleic acids d. Enzymes
4. Which of the following is not a strong electrolyte
   a. $CH_3COOH$ b. $NaClO_4$ c. KCl d. $KNO_3$
5. Electrode potential is the
   a. Potential difference between electrode and electrolyte surfaces
   b. difference between electrode and metal
   c. difference between anode and cathode
   d. None of the above
6. DHO theory applicable to
   a. Strong electrolyte b. Weak electrolyte
   c. both strong and weak electrolyte d. None of the above
7. Internal conversion is the transition involve
   a. Higher excited state to ground state b. Ground state to first excited state
   c. Higher excited state to first excited state d. Ground state to higher excited state
8. Mathematical expression of Stern-Volmer equation is
   a. 1 - K[Q] b. 1-Ksv[Q] c. 1+Ksv[Q] d. Ksv[Q}
9. Excimer is formed from
   a. Monomers of same species b. Monomers of different species
   c. Monomer with photosensitizer d. None of the above
10. Photo chemistry deals with the study of
    a. Photo
    b. Photos
    c. Reactions which proceed with absorptions of UV light
    d. Reactions which proceed with absorption of IR light

**II. Match the following**

11. Enzyme - ø
12. Excimer - Catalyst
13. Probability factor - H
14. Enthalpy - ρ
15. Quantum efficiency - Dimer

## SECTION – B (10X2=20)

Answer any **TEN** Questions

16. Define collision cross section?
17. What is the range of value of Probability factor?
18. How does the pressure affect the rate of reaction in solution reaction?
19. Write the mathematical form of Bronsted catalysis law.
20. Define quantum efficiency.
21. Define ionic strength?
22. Define autocatalysis.
23. Define electrode potential of metal.
24. What is zwitter ions?
25. What is the meaning of flux?
26. What is meant by linear forces?
27. What is meant by triplet state?

## SECTION- C (5X8=40)

Answer any **FIVE** Questions

28. a) What is meant by competitive inhibition of enzyme? **(2)**
    b) What is opposing reaction? Give one example. **(3)**
    c) Write a note on radiolysis of water. **(3)**

29. a) Define chain length. **(2)**
    b) Write equation for decomposition of Nitrogen pentoxide. **(2)**
    c) Write short notes on influence of pH of the medium on enzyme catalysis. **(4)**

30. a) Mention two types of adsorption. **(2)**
    b) What is intrinsic semiconductor? **(2)**
    c) Explain stopped flow and flash photolysis methods in fast reactions. **(4)**

31. a) What is fluorescence? **(2)**
    b) Define the internal conversion? **(2)**
    c) What is photosensitization? Give one example? **(4)**

32. a) Write the difference between physisorption and chemisorption. **(3)**
    b) Discuss the kinetics of consecutive reactions. **(5)**

33. a) Write the butler – Volmer equation for single step reaction and mention the terms. **(4)**
    b) What is electric field force? **(2)**
    c) What is mean by antistokes shift? **(2)**

34. a) Asample of Gaseous HI was irradiated by light of wavelength 253.7 nm when 307 J of energy was found to decompose $1.30 \times 10^{-3}$ mole of HI.Calculate the quamtum yield for the dissociation of HI. **(4)**
    b) Discuss Helmholtz double layer model. **(4)**

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